

Supplementary Materials for the paper “Beyond Visual Recognition: Evaluating Cultural-Semiotic Reasoning in Slovenian Vision-Language Models”

Anonymous TACL submission

A Limitations

The benchmark and evaluation have several limitations.

First, the reference answers were generated by Gemini-2.5-Flash rather than independently written by human experts. Contextual text improves factual specificity but can also introduce errors when the text is only weakly connected to the image. Although the benchmark includes text–image correlation metadata, automatically generated answers should be regarded as silver-standard references.

Second, the same model family contributes to several stages of the pipeline. Gemini-2.5-Flash was used for image classification, search-prompt generation, secondary image filtering, question suitability assessment, and reference-answer generation, while Gemini-2.5-Pro was used as the final judge. This creates the possibility of shared stylistic or architectural bias. The judge may prefer interpretations that resemble the reference model’s outputs even when alternative answers are equally valid.

Human validation was limited to 50 of the 4,090 evaluated image–question pairs, corresponding to approximately 1.2% of the benchmark instances. Although the sample was balanced across the four broad task groups, it was not balanced across all ten individual question types, and some subquestions were represented by only one instance. Category-level agreement estimates should therefore be interpreted as exploratory.

The two human annotators agreed on the best answer in 42% of cases, and agreement between the automatic judge and individual annotators ranged from 34% to 48%. Gemini aligned more strongly with cases on which the humans reached consensus, but it still disagreed with the consensus answer in nine of 21 cases. Moreover, the human model-preference distributions differed from the

automatic ranking. These findings demonstrate that selecting a single best answer for open-ended benchmark questions is partly subjective and that the reported model ordering is dependent on the chosen evaluator.

Third, category assignment was largely automated. Images could receive incorrect or overly broad OCM labels, and a top-level category may contain visually and conceptually diverse material. The use of top-level categories improves coverage but reduces taxonomic precision.

Fourth, the benchmark contains only five deliberately sampled images per category. Although multi-label overlap increases the observed category counts, the number of unique examples for rare cultural domains remains small. The benchmark is therefore more suitable for broad diagnostic comparison than for estimating performance within every individual OCM category.

Fifth, some categories were difficult to populate. Sexuality and reproduction were particularly constrained by search-engine safety systems, ethical considerations, and the limited availability of culturally specific non-explicit photographs. The corresponding samples may not capture the full conceptual breadth of these categories.

The fine-tuning experiment introduces additional limitations. The training targets are silver-standard answers generated through the same general model family used elsewhere in the benchmark pipeline. The adapter may consequently learn the wording, structure, or interpretive preferences of the reference generator. Because Gemini-2.5-Pro also served as the pairwise judge, stylistic alignment may contribute to the observed score improvement even though candidate identities were hidden.

The pairwise SFT comparison was evaluated by one automated judge and was not independently repeated by the human annotators. It used one training seed, one epoch, and one QLoRA configuration,

without comparison against full-parameter fine-tuning, vision-side adaptation, alternative LoRA ranks, or different training-data filters. The training distribution was also uneven across question types because questions judged unsuitable were removed during data preparation. These results should therefore be interpreted as evidence about this specific parameter-efficient adaptation procedure rather than as a general estimate of the maximum benefit obtainable from the dataset.

Finally, absolute scores from the pairwise SFT evaluation should not be compared directly with the scores from the original five-model evaluation. Although both evaluations use the same rubric, the number and identity of candidate answers presented to the judge differ. Conclusions about fine-tuning should therefore be based on the paired base-adapter differences within the dedicated comparison.

Finally, the study evaluates only models with at most 12 billion parameters. Larger proprietary and open-weight VLMs may display substantially stronger performance on the benchmark’s semiotic reasoning tasks. The results therefore characterise compact and moderately sized models rather than the full state of the art.

B Ethical Considerations

The use of cultural categories can unintentionally reinforce essentialist assumptions. OCM labels describe analytical domains rather than fixed properties of Slovenian identity. An image assigned to a category should not be interpreted as a complete or authoritative representation of Slovenian culture.

Sensitive categories, including sexuality, reproduction, illness, death, religion, crime, and social problems, require additional care. The collection process excluded explicit, exploitative, graphic, or personally identifying material.

C Related Work

C.1 Vision-Language Models

Modern VLMs integrate a visual encoder with a language model in order to process images and textual instructions jointly. Early contrastive systems such as CLIP learn aligned image and text representations, while later generative architectures connect visual encoders to autoregressive language models and support open-ended multimodal interaction (Radford et al., 2021; Alayrac et al., 2022;

Liu et al., 2023). Recent model families, including Gemma, Qwen-VL, and other multilingual VLMs, have improved visual instruction following, multilingual communication, and reasoning over high-resolution images (Wang et al., 2024; Gemma Team, 2025).

The present study focuses on compact models with at most 12 billion parameters. This restriction reflects an interest in systems that can be evaluated or deployed with moderate computational resources. It also enables a direct comparison between generic multilingual models and models adapted specifically for Slovenian.

C.2 Multimodal Evaluation

Common VLM benchmarks assess tasks such as image captioning, object recognition, OCR, visual grounding, and visual question answering. Representative examples include VQAv2, which evaluates visual question answering, and TextVQA, which evaluates reasoning over text embedded in images (Goyal et al., 2017; Singh et al., 2019). Such benchmarks commonly rely on answer accuracy, exact or approximate agreement with reference answers, and related automatic metrics. These measures are effective for short and relatively objective answers, but are less suitable for open-ended explanations that may be semantically valid despite differing substantially in wording.

LLM-as-a-judge evaluation has consequently become a common alternative for assessing generated responses (Chiang and Lee, 2023; Zheng et al., 2023). In this setting, a capable language or multimodal model assigns scores according to predefined criteria or selects the best response among several candidates. Although this approach supports flexible semantic evaluation, it can introduce evaluator bias, positional bias, preference for specific response styles, and sensitivity to prompt design. Human validation is therefore important when an LLM judge is used as the principal evaluation method.

C.3 Semiotics and Visual Meaning

Semiotics studies how signs produce and communicate meaning. In visual communication, meaning is not limited to the identification of depicted objects. It also emerges from colour, framing, composition, visual hierarchy, social conventions, symbols, metaphorical associations, and relations between visual and textual modes (Kress and van Leeuwen, 2021; Stöckl, 2024). For the purposes of

this benchmark, the distinction between denotation and connotation can be understood operationally as a distinction between directly represented content and culturally mediated or associative meaning (Chandler, 2022). These dimensions need not be understood as fixed or mutually exclusive stages of semiosis: plausible interpretations may vary, while remaining constrained by visual evidence, cultural conventions, and communicative context. Throughout this paper, we use ‘cultural-semiotic reasoning’ as an operational term for tasks that require models to relate visible features to compositional organisation, symbolic or metaphorical associations, intertextual references, communicative intent, and culturally situated meaning. Here, ‘multimodal’ primarily characterises the model architecture and image–language setting; the benchmark itself focuses more specifically on visual meaning articulated through language rather than on multimodal artefacts in which several semiotic modes jointly organise meaning.

Several existing datasets address individual aspects of higher-level visual interpretation. The advertisement-understanding dataset introduced by Hussain et al. (2017) includes persuasive intent, symbolic references, and the actions encouraged by advertisements. ArtEmis connects artworks with emotional reactions and natural-language explanations that may refer to abstract concepts, metaphors, and subjective associations (Achlioptas et al., 2021). Most directly related to the present work, the *Impressions* dataset collects annotations of pragmatic image description, aesthetic impact, viewer impressions, and visual design choices (Kruk et al., 2023). Recent work has also begun to operationalise concepts from visual semiotics computationally, illustrating how semiotically informed distinctions can be translated into structured image-analysis frameworks (Morra et al., 2024). However, these resources are predominantly English-language and domain-specific, and they do not jointly evaluate culturally situated reasoning across the range of semiotic tasks considered here.

The present benchmark adopts a broader task-based perspective and distinguishes three task groups of semiotic reasoning. Foundational tasks concern perceptual organisation and visual structure. Intermediate tasks concern composition, symbols, and metaphors. Advanced tasks concern connotation, intertextuality, communicative intent, and audience. These labels denote progressively more

interpretive forms of visual reasoning rather than an empirically established hierarchy of task difficulty.

C.4 Slovenian Vision-Language Resources

Slovenian is a less-resourced South Slavic language. Recent work by the GaMS research group introduced a large Slovenian multimodal instruction-tuning dataset and the SVILA family of vision-language models (Martinc and Vreš, 2026). SVILA-1-4B and SVILA-1-12B were obtained by fine-tuning the corresponding multilingual Gemma-3 models on more than one million Slovenian image–text instruction pairs. Their evaluation demonstrated improvements in Slovenian visual question answering, visual grounding, optical character recognition, and instruction following.

The present work examines a complementary question: whether Slovenian adaptation also improves culturally situated semiotic reasoning. Unlike conventional VQA benchmarks, our questions require models to explain symbolism, composition, metaphor, connotation, and cultural relevance.

To the best of our knowledge, no previous benchmark has been designed specifically to evaluate visual semiotic reasoning in Slovenian. Existing Slovenian VLM evaluations primarily cover general visual question answering, OCR, visual grounding, and instruction following (Martinc and Vreš, 2026), while related datasets targeting visual-semiotic interpretation have focused on English-language and domain-specific material (Kruk et al., 2023).

D QLoRA Configuration

We used quantised low-rank adaptation, or QLoRA (Hu et al., 2022; Dettmers et al., 2023). The pretrained model weights were loaded using 4-bit NormalFloat quantisation, double quantisation, and bfloat16 computation. Quantisation reduced memory requirements but did not convert the procedure into dense 4-bit fine-tuning: the quantised pretrained weights remained frozen, and optimizer updates were applied only to the LoRA parameters.

For a pretrained linear transformation with weight matrix W , LoRA introduces a low-rank correction:

$$y = Wx + \frac{\alpha}{r}BAx, \quad W_{\text{effective}} = W + \frac{\alpha}{r}BA.$$

We used rank $r = 16$, scaling parameter $\alpha = 32$, and therefore a correction scale of $\alpha/r = 2$.

LoRA dropout was set to 0.05, and no existing bias parameters were updated.

Adapters were attached only to the following projections within the language transformer:

- attention projections: q_proj, k_proj, v_proj, and o_proj;
- MLP projections: gate_proj, up_proj, and down_proj.

The targeting rule was restricted explicitly to modules under `model.language_model`. The vision encoder and multimodal projector therefore received no LoRA adapters.

The configuration introduced 65,470,464 trainable LoRA parameters, approximately 0.5% of the nominal 12-billion-parameter backbone. The experiment therefore adapted a comparatively small language-side parameter set rather than directly updating the complete model.

Although the visual encoder was frozen, the language-side adapters could still change how image-derived representations were attended to and combined with Slovenian question tokens. They could not, however, change the initial extraction of visual features or the projector that maps those features into the language-model representation space.

E Training Configuration

Training was performed for one epoch on two NVIDIA A100 40 GB GPUs. The per-device batch size was one, with four gradient-accumulation steps, producing an effective global batch size of eight. This resulted in 4,368 optimizer steps.

Gradient checkpointing was enabled, while sequence packing and image pan-and-scan processing were disabled. The completed run obtained an average training loss of 1.059. At the end of the epoch, validation loss was 1.023 and validation mean token accuracy was 0.728. The saved output consists of the LoRA adapter and processor configuration; the original Gemma-3-12B-IT checkpoint remains necessary when loading the adapted model.

F All-Pairs Score Sums by Evaluation Dimension

Table 3 reports the all-pairs score sums for task fulfilment, visual grounding, and explanatory coherence. Gemma-3-12B-IT obtains the highest all-pairs score on all four dimensions. EuroVLM-9B

nearly matches it in hallucination control, despite producing fewer responses labelled as major hallucinations. These measures are related but distinct: hallucination control is an ordinal score assigned to every answer, whereas major hallucination is a separate binary label.

Among the 4B models, Gemma-3-4B-IT obtains higher fulfilment and visual grounding, while SVILA-1-4B obtains slightly higher explanation and substantially higher hallucination control scores.

G Performance by Question Type

Gemma-3-12B-IT achieves the highest aggregate score for every question type in the all-pairs evaluation. Its strongest absolute scores occur for F3, concerning colour semantics and mood; I1, concerning composition; F2, concerning object relations; and A3, concerning intent and audience. The relative difficulty of tasks cannot be inferred directly from raw score sums because the number of valid instances differs slightly between question types. Nevertheless, several recurring tendencies are visible.

First, intertextual understanding, corresponding to A2, produces lower scores for most models. This task often requires knowledge of specific artworks, historical narratives, genres, or cultural references that are not visually explicit. Second, the SLO question is difficult for the generic 4B model, suggesting that connecting an image with Slovenian cultural meaning requires either stronger world knowledge or better Slovenian-language adaptation. Third, foundational tasks are not uniformly easier than advanced tasks. Questions about spatial relations and visual salience can require precise image grounding, while advanced questions sometimes permit broader but plausible interpretations.

Table 6 reports best-answer selections by question type.

The Slovenian-adapted SVILA models are comparatively competitive on the SLO question. SVILA-1-12B is selected 101 times, compared with 73 selections for EuroVLM-9B, while SVILA-1-4B is selected 69 times, considerably more often than Gemma-3-4B-IT. This provides some evidence that Slovenian adaptation benefits explicitly cultural questions. However, this advantage is not sufficient to offset weaker performance across the remaining semiotic tasks.

Component	Function	Updated during SFT?
Vision encoder	Converts image patches into visual features	No
Multimodal projector	Maps visual features into language-model space	No
Token embeddings	Converts text tokens into representations	No
Dense attention weights	Performs pretrained attention operations	No
Dense MLP weights	Performs pretrained representation processing	No
LoRA attention adapters	Modifies language-side attention behaviour	Yes
LoRA MLP adapters	Modifies language-side feature processing	Yes
Output language-model head	Maps representations to token probabilities	No
Existing bias vectors	Bias terms in pretrained transformations	No

Table 1: Frozen and trainable components in the QLoRA experiment.

Hyperparameter	Value
Training examples	34,944
Validation examples	3,924
Epochs	1
Effective global batch size	8
Optimizer	Fused AdamW
Peak learning rate	1×10^{-4}
Learning-rate schedule	Cosine
Warm-up ratio	0.03
Weight decay	0.01
LoRA rank	16
LoRA alpha	32
LoRA dropout	0.05
Quantisation	4-bit NF4
Computation	bfloat16
Random seed	42

Table 2: Configuration of the supervised fine-tuning experiment.

H Comparison between Gemma-3-12B-IT and its QLoRA adapter across evaluation dimensions and question type

Table 7 presents the controlled comparison between Gemma-3-12B-IT and its QLoRA adapter across different evaluation dimensions. The improvement was not uniform across the four dimensions. Table 7 shows that the largest gain occurred in hallucination control, followed by task fulfilment and explanation. Visual grounding decreased slightly in the all-pairs evaluation and more clearly in the question-suitable subset.

The score distribution indicates that SFT primarily raised the lower end of model performance. The proportion of responses scoring at most two points decreased from 19.7% to 8.1%, while the proportion scoring at least six increased from 59.6% to 68.5%. However, the proportion receiving the maximum score of eight decreased from 36.1% to 32.9%. The adapter therefore produced fewer

catastrophic failures and more consistently adequate answers, but also fewer responses receiving a perfect score.

The effect of fine-tuning also differed across question types, as shown in Table 8. The largest improvements occurred for intertextual understanding (A2), symbol recognition in cultural context (I2), and Slovenian cultural relevance (SLO). A3 was the only question type for which the total score decreased.

When the question types are combined into broad families, the mean total score improves by 0.652 points for advanced questions, 0.341 for foundational questions, 0.575 for intermediate questions, and 1.054 for the SLO question. Nevertheless, direct best-answer rates exceed 50% only for A2, I2, and SLO. This again shows that higher average rubric scores should not be treated as equivalent to a uniform preference for the adapted responses.

I Literal Recognition versus Meaning Interpretation

The task-level results support the premise that detailed image descriptions do not guarantee culturally grounded interpretation. Models often identify major objects and produce plausible emotional language, but they struggle to connect visible details with specific cultural or intertextual meanings.

This difficulty is particularly evident in intertextual questions. A valid answer may require recognition of a known artwork, a national narrative, a religious motif, a historical event, or a visual genre. When such a reference is uncertain, models may either produce an overly generic answer or hallucinate a specific source. Similarly, metaphor questions encourage interpretation but increase the risk of unsupported symbolism.

The benchmark therefore exposes a central ten-

Model	Fulfilment	Visual grounding	Explanation	Hallucination control
Gemma-3-12B-IT	6,586	5,879	6,195	5,589
EuroVLM-9B	5,897	4,406	4,784	5,579
SVILA-1-12B	5,464	4,532	4,585	5,387
SVILA-1-4B	4,485	3,290	3,401	5,024
Gemma-3-4B-IT	4,882	3,367	3,277	3,098

Table 3: All-pairs score sums across the four evaluation dimensions. Each individual score ranges from zero to two. For every model, the four dimension sums add to its overall total score.

Category	EuroVLM-9B	SVILA-1-12B	SVILA-1-4B	Gemma-3-12B-IT	Gemma-3-4B-IT
F	6,589	5,978	4,960	8,177	4,834
I	6,471	5,813	4,627	7,263	4,575
A	5,969	6,367	4,974	7,115	4,298
SLO	1,637	1,810	1,639	1,694	917

Table 4: All-pairs score sums aggregated by question family. F combines F1–F3, I combines I1–I3, and A combines A1–A3.

sion in the evaluation of semiotic reasoning. A model should provide an interpretation rich enough to go beyond literal description, yet it must remain grounded in visual evidence and appropriately express uncertainty. The best-performing models are those that balance these requirements rather than maximising either descriptiveness or caution alone.

J Role of the Slovenian Cultural-Relevance Question

The SLO question directly asks what an image means for Slovenia. It differs from the other tasks because answering it often requires information not visually recoverable from the image. A landscape may be recognisable as Slovenian only through knowledge of a specific mountain, river, or settlement. An artwork may require knowledge of its author or place in Slovenian art history. A ritual or object may have a particular national or regional significance.

The effect of Slovenian adaptation on the SLO question is scale-dependent. SVILA-1-4B was selected as the best answer 69 times, compared with 25 selections for Gemma-3-4B-IT, indicating a clear benefit at the smaller scale. In contrast, SVILA-1-12B was selected 101 times, compared with 141 selections for Gemma-3-12B-IT. Because the SVILA models use the corresponding Gemma models as their backbones, the 12B result suggests that the adaptation procedure did not preserve all capabilities relevant to cultural-semiotic interpretation. Overfitting to the instruction-tuning distribution or degradation of more general instruction-following abilities are plausible explanations, al-

though the present experiment cannot determine the cause.

Reference construction for SLO questions depends strongly on the quality and specificity of the contextual source text. In the 13 SLO instances included in human validation, the two annotators agreed on the best answer in four cases, and Gemini matched the human consensus in two of those cases. Although the subset is too small for firm conclusions, it illustrates the difficulty of adjudicating culturally specific interpretations and supports reporting SLO results with appropriate caution.

K Thorough Analysis of Human Validation and Inter-Annotator Agreement

Table 9 summarises the human-validation results. The two annotators selected the same best answer in 21 of 50 instances, corresponding to 42.0% raw agreement and Cohen’s $\kappa = 0.249$. The chance-corrected agreement is therefore limited, illustrating the subjectivity of selecting a single best response for open-ended benchmark questions.

Annotator 1 agreed with Gemini’s best-answer selection in 17 of 50 instances (34.0%; $\kappa = 0.156$), whereas Annotator 2 agreed with Gemini in 24 instances (48.0%; $\kappa = 0.327$). Gemini therefore aligned more closely with Annotator 2 than with Annotator 1. Across the 21 instances on which the two humans agreed, Gemini selected the same answer in 12 cases, yielding 57.1% raw agreement and $\kappa = 0.439$. The higher agreement on this consensus subset suggests that the automatic judge is more reliable when the preferred answer is compar-

Category	EuroVLM-9B	SVILA-1-12B	SVILA-1-4B	Gemma-3-12B-IT	Gemma-3-4B-IT
F	6,220	5,669	4,719	7,684	4,574
I	5,351	4,803	3,927	6,094	3,912
A	4,727	5,000	4,007	5,886	3,688

Table 5: Question-suitable score sums aggregated by question family. F combines F1–F3, I combines I1–I3, and A combines A1–A3.

Model	A1	A2	A3	F1	F2	F3	I1	I2	I3	SLO
Gemma-3-12B-IT	218	146	263	236	255	276	261	189	233	141
SVILA-1-12B	76	108	60	60	51	49	54	81	75	101
EuroVLM-9B	40	76	32	65	53	25	40	64	40	73
Gemma-3-4B-IT	38	35	31	25	31	37	41	38	39	25
SVILA-1-4B	37	44	23	23	19	22	13	37	22	69

Table 6: Number of best-answer selections in the all-pairs evaluation, separated by question type.

Dimension	All-pairs setting 4,090			Question-suitable subset 3,118		
	Base	SFT	Difference	Base	SFT	Difference
Task fulfilment	1.528	1.662	+0.134	1.616	1.686	+0.070
Visual grounding	1.322	1.299	-0.023	1.413	1.333	-0.080
Explanation	1.416	1.526	+0.110	1.497	1.538	+0.041
Hallucination control	1.250	1.604	+0.354	1.322	1.600	+0.278
Total	5.516	6.091	+0.576	5.849	6.158	+0.309

Table 7: Mean scores before and after supervised fine-tuning. Each dimension ranges from zero to two, and the total ranges from zero to eight.

Question type	Base mean	SFT mean	Difference	SFT best-answer rate
A1	5.396	5.619	+0.222	44.7%
A2	4.465	6.328	+1.863	64.1%
A3	6.242	6.112	-0.130	38.9%
F1	5.963	6.169	+0.205	43.3%
F2	5.902	6.352	+0.450	47.9%
F3	6.347	6.714	+0.367	47.7%
I1	6.161	6.318	+0.156	42.3%
I2	4.604	6.007	+1.403	58.7%
I3	5.919	6.083	+0.164	41.6%
SLO	4.156	5.210	+1.054	50.9%

Table 8: Pairwise fine-tuning results by question type.

atively unambiguous. Nevertheless, disagreement remained in nine of the 21 consensus cases, so these results should not be treated as a complete validation of the automatic judge.

The three-way Fleiss' κ for Annotator 1, Annotator 2, and Gemini was 0.231. This value should be interpreted as three-rater agreement rather than purely human inter-annotator agreement because one of the three raters was an automatic judge.

The mean question-relevance rating was 3.45 out of five. The individual annotator means were nearly identical: 3.46 for Annotator 1 and 3.44 for Annotator 2. Exact agreement was only 32.0%, but the annotators differed by at most one scale point in 41 of 50 cases (82.0%). Their mean absolute difference was 0.90 points. Linearly weighted κ was 0.391, while quadratically weighted κ was 0.598. These results indicate that the annotators often differed in rating intensity but generally agreed on the broader level of question relevance.

A manual inspection of the instances where the annotators differed by more than one scale point in their question-relevance rating further demonstrates that a culturally grounded (human) interpretation of an image can vary across viewers, particularly for open-ended questions. A question about the predominant colors of the image and the emotions these evoke may to some people seem irrelevant if the actual content of the image is evocative in some other meaningful way, e.g. by people occupying different spaces in a large gallery with a prominently symbolic monument. Similarly, asking in what way an image of a wedding expresses the meaning of Slovenia may evoke differing responses depending on the values and beliefs one associates with marriage on the one hand and national identity on the other.

Silver-standard reference answers received a higher pooled mean rating of 4.36. The annotator-specific means were 4.32 and 4.40. Exact agreement reached 52.0%, and the ratings differed by at most one point in 43 of 50 cases (86.0%). The mean absolute difference was 0.64 points. Linearly weighted κ was 0.264 and quadratically weighted κ was 0.369.

The relatively low chance-corrected agreement for reference quality should be considered together with the strong concentration of ratings at the upper end of the scale: 83% of all reference-quality ratings were either four or five. This ceiling effect increases expected chance agreement and can

depress κ even when raw agreement is comparatively high. The results therefore indicate that the automatically generated references were generally considered good, but they should still be treated as silver-standard rather than expert-verified gold-standard answers.

Human agreement varied across the four broad question groups. Agreement on the best answer was highest for advanced questions, with 58.3% raw agreement and $\kappa = 0.450$ across 12 instances. It was lower for intermediate questions (41.7%; $\kappa = 0.192$), foundational questions (38.5%; $\kappa = 0.181$), and Slovenian cultural-relevance questions (30.8%; $\kappa = 0.152$).

The particularly low agreement for the SLO group is consistent with the open-ended and culturally specific nature of these questions. Different answers may emphasise national identity, regional significance, historical context, or visible evidence while remaining partially defensible. The comparatively high agreement for the advanced group should not be interpreted as evidence that advanced questions are inherently easier to evaluate, because each category contains only 12 or 13 observations and the category-level estimates are consequently unstable.

Within the human-consensus subset, Gemini matched the agreed human answer in four of five foundational cases, four of seven advanced cases, two of five intermediate cases, and two of four SLO cases. These subgroup counts are too small for firm comparisons, but they suggest that the judge's alignment with human consensus varies across task types.

L OCM Category Classification Prompt

L.1 Original Slovenian Prompt

Opravi klasifikacijo slike po kategorijah OCM.

Navodila:

- 1) Izberi VSE ustrezne kategorije iz spodnjega seznama.
- 2) Uporabi IZKLJUČNO kategorije iz seznama (ne izmišljaj novih).
- 3) Če nobena kategorija ne ustreza, vrni prazen seznam [].
- 4) Odgovori IZKLJUČNO kot veljaven JSON seznam nizov, npr. ["geografija", "zgodovina in kulturne spremenbe"].

Seznam dovoljenih (vrhnjih) kategorij:
{cats}

L.2 English Translation

Classify the image according to the OCM categories.

Instructions:

Assessment or comparison	n	Mean	Exact agreement	κ	Statistic
Question relevance	50	3.45	0.320	0.391	Linear weighted
Reference-answer quality	50	4.36	0.520	0.264	Linear weighted
Annotator 1 vs. Annotator 2, best answer	50	—	0.420	0.249	Unweighted
Annotator 1 vs. Gemini, best answer	50	—	0.340	0.156	Unweighted
Annotator 2 vs. Gemini, best answer	50	—	0.480	0.327	Unweighted
Human consensus vs. Gemini, best answer	21	—	0.571	0.439	Unweighted

Table 9: Human-validation results. Means pool the ratings of both annotators. Linear weighting is used for the two five-point ordinal scales; unweighted Cohen’s κ is used for nominal best-answer selections.

Model	Ann. 1	Ann. 2	Gemini
EuroVLM-9B	19	12	11
Gemma-3-12B-IT	9	12	22
Gemma-3-4B-IT	6	1	5
SVILA-1-12B	11	17	8
SVILA-1-4B	5	8	4

Table 10: Best-answer selections in the 50-instance human-validation sample. Gemini results correspond to the same exact image–question pairs evaluated by the humans.

- 1) Select ALL relevant categories from the list below.
- 2) Use ONLY categories from the provided list (do not invent new categories).
- 3) If none of the categories applies, return an empty list [].
- 4) Respond ONLY with a valid JSON list of strings, for example:
["geografija",
"zgodovina in kulturne spremenbe"].

List of permitted top-level categories:
{cats}

M Targeted Search-Prompt Generation

M.1 Original Slovenian Prompt

SISTEMSKO NAVODILO:

Si generator iskalnih pozivov (promptov) v slovenščini za zbiranje fotografij, kulturno povezanih s Slovenijo. Pozivi naj bodo konkretni in primerni za Bing Image Search (realne fotografije). Izogibaj se politični propagandi, logotipom in avtorsko zaščitanim likom. Vrni SAMO veljaven JSON.

UPORABNIŠKO NAVODILO:

Kategorija: {category_label}

Ustvari {n} različnih iskalnih pozivov v slovenščini, kulturno povezanih s Slovenijo. Vsak prompt naj bo primeren za iskanje fotografij kraja, dogodka, predmeta ali običaja.

Izhodna JSON shema:

```
{
  "items": [
    {
      "prompt": "slovenski iskalni niz",
```

```
      "description": "Kratek opis kulturnega pomena."
    }
  ]
}
```

M.2 English Translation

SYSTEM INSTRUCTION:

You are a generator of Slovenian-language search queries for collecting photographs that are culturally connected to Slovenia. The queries should be concrete and suitable for Bing Image Search, with a focus on real photographs. Avoid political propaganda, logos, and copyrighted fictional characters. Return ONLY valid JSON.

USER INSTRUCTION:

Category: {category_label}

Generate {n} different Slovenian-language search queries that are culturally connected to Slovenia. Each query should be suitable for finding photographs of a place, event, object, or custom.

Output JSON schema:

```
{
  "items": [
    {
      "prompt": "Slovenian search query",
      "description": "A short description of its cultural significance."
    }
  ]
}
```

N Reference-Answer Generation Prompt

N.1 Original Slovenian Prompt

Odgovori na vprašanja o tej sliki, pri tem pa upoštevaj tudi dodatno besedilo: {text}

Vrni slovar z naslednjimi vrhnjimi ključi:
{fields_text}

Dosledno upoštevaj naslednjo strukturo in jezikovne zahteve:

- Odgovori morajo biti v slovenščini.
- Besedila ne omenjaj neposredno, ampak ga uporabi za oblikovanje pravih odgovorov.
- Pod zahtevanimi ključi zapiši samo vprašanje.
- Pod suitable presodi, kako primerno je vprašanje za

- sliko, in zapiši da, srednje ali ne.
- Pod `text_image_correlation` presodi, kako primerno je besedilo za sliko, in zapiši dobro, srednje ali slabo.
 - Vrni IZKLJUČNO veljaven JSON.

N.2 English Translation

Answer the questions about this image while also taking the following additional text into account: {text}

Return a dictionary with the following top-level keys: {fields_text}

Strictly follow the required structure and language requirements:

- The answers must be written in Slovenian.
- Do not refer to the additional text explicitly. Use it only to formulate accurate answers.
- Under the required keys, include only the corresponding question.
- Under suitable, assess how appropriate the question is for the image and return yes, medium, or no.
- Under `text_image_correlation`, assess how well the text corresponds to the image and return good, medium, or poor.
- Return ONLY valid JSON.

O LLM-as-a-Judge Prompt

O.1 Original Slovenian Prompt

The automatic judge was prompted in Slovenian. The prompt is reproduced verbatim below, followed by an English translation. Although the benchmark answers are described in this paper as silver-standard answers, the prompt itself used the Slovenian term *zlati odgovor* ("gold answer"). We preserve this terminology here to reproduce the actual evaluation procedure accurately.

O.2 Original Slovenian Prompt

DIM_DEFINITIONS = Ocenjuješ kakovost odgovora LLM glede na SLIKO, VPRAŠANJE in ZLATI ODGOVOR.

Imaš 4 SKUPNE DIMENZIJE (vsaka 0-2 točki):

- 1) Izpolnitev naloge (fulfillment, 0-2)
 - 0 - zgreši bistvo ali odgovori na drugo temo.
 - 1 - delno; manjkajo ključni deli ali so preohlapni.
 - 2 - v celoti odgovori na zahtevo vprašanja.
- 2) Vizualna utemeljenost & specifičnost (visual_grounding, 0-2)
 - 0 - generičen opis, sklicevanje na nevidne stvari.
 - 1 - nekaj konkretnih sklicev na vidne elemente.
 - 2 - jasne, specifične podrobnosti (položaj, velikost, barve, prekrivanje, kompozicija, ipd.).
- 3) Razlaga & koherenca (coherence, 0-2)
 - 0 - nelogično, protislovno, skokovito.
 - 1 - razumljivo, a tanko ali deloma nedosledno.
 - 2 - jasna, povezovalna razlaga, ki podpira trditve.

- 4) Nadzor halucinacij (hallucination, 0-2)
 - 0 - izmišlja si glavne objekte/odnose/barve (velika halucinacija).
 - 1 - manjše pretiravanje ali ugibanje brez označene negotovosti.
 - 2 - brez halucinacij glede slike; negotovost po potrebi jasno označena.

Pomembno:

- Zlati odgovor je referenca, kaj je dober odgovor, vendar naj bo primarni kriterij skladnost s sliko in vprašanjem.
- Parafraziranje v modelovem odgovoru je OK, dokler vsebinsko ustreza.
- Ne izmišljaj neobstoječih objektov, barv ali simbolov.

Deluješ kot VLM sodnik. Ocenjuješ odgovor LLM na ENO specifično vizualno nalogo.

Najprej preberi definicije 4 dimenzij ocenjevanja: {DIM_DEFINITIONS}

Na voljo imaš:

- sliko (kot vizualni input),
- vprašanje (Question),
- zlati odgovor (Gold_answer),
- modelov odgovor (Model_answer), ki ga ocenjuješ.

Podatki:

Question: {q_text}
Gold_answer: {gold_answer}
Model_answer: {model_answer}

Tvoja naloga:

Oceniti Model_answer na 4 dimenzijah. Vsaka dimenzija mora biti celo število 0, 1 ali 2.

Dimenzije izhoda:

- fulfillment
- visual_grounding
- coherence
- hallucination

IZHOD:

Vrni IZKLJUČNO en JSON objekt brez dodatne razlage v natančno tej obliki:

```
{{
  "fulfillment": 0,
  "visual_grounding": 0,
  "coherence": 0,
  "hallucination": 0
}}
```

Seveda pa vrednosti 0 zamenjaj z dejanskimi ocenami 0, 1 ali 2.

Ne dodajaj nobenih dodatnih polj ali besedila.

O.3 English Translation

You evaluate the quality of an LLM response against the IMAGE, QUESTION, and GOLD ANSWER.

There are 4 SHARED DIMENSIONS (each scored from 0 to 2):

1. Task fulfilment (fulfillment, 0-2)
 - 0 - Misses the main point or addresses a different topic.

- 1 – Partial; key elements are missing or treated too vaguely.
2 – Fully addresses the requirements of the question.
2. Visual grounding & specificity (visual_grounding, 0-2)
0 – Generic description or references to things that are not visible.
1 – Some concrete references to visible elements.
2 – Clear, specific details (position, size, colours, overlap, composition, etc.).
3. Explanation & coherence (coherence, 0-2)
0 – Illogical, contradictory, or disjointed.
1 – Understandable but superficial or partly inconsistent.
2 – A clear, coherent explanation that supports its claims.
4. Hallucination control (hallucination, 0-2)
0 – Invents major objects, relationships, or colours (major hallucination).
1 – Minor exaggeration or speculation without explicitly indicating uncertainty.
2 – No hallucinations about the image; uncertainty is clearly indicated where appropriate.

Important:

- * The gold answer serves as a reference for what constitutes a good answer, but consistency with the image and question should be the primary criterion.
- * Paraphrasing in the model's response is acceptable as long as the content remains appropriate.
- * Do not invent objects, colours, or symbols that are not present.

You act as a VLM judge. You evaluate an LLM response to ONE specific visual task.

First, read the definitions of the four evaluation dimensions:
{DIM_DEFINITIONS}

You are provided with:

- * an image (as visual input),
- * a question (Question),
- * a gold answer (Gold_answer),
- * the model response being evaluated (Model_answer).

Data:

Question: {q_text}
Gold_answer: {gold_answer}
Model_answer: {model_answer}

Your task:

Evaluate Model_answer on four dimensions. Each dimension must be assigned an integer score of 0, 1, or 2.

Output dimensions:

- * fulfillment
- * visual_grounding
- * coherence
- * hallucination

OUTPUT:

Return EXCLUSIVELY one JSON object, without any additional explanation, in exactly the following format:

```
{{
  "fulfillment": 0,
  "visual_grounding": 0,
  "coherence": 0,
  "hallucination": 0
}}
```

Replace the values of 0 with the actual scores of 0, 1, or 2.

Do not add any additional fields or text.

P Instructions to Human Annotators

P.1 Original Slovenian Instructions

TVOJA NALOGA:

Oglej si sliko, preberi vprašanje in referenčni odgovor. Sledi 5 kandidatnih odgovorov, ki so jih napisali različni modeli. Tvoje naloge so tri:

I. Ocena relevantnosti vprašanja za sliko (ocena od 1 do 5):

- 5: Vprašanje je relevantno, se povezuje s sliko in sprašuje po specifikah ali podrobnostih, ki jih je na sliki mogoče opaziti.
- 4: Vprašanje je delno relevantno, se v splošnem lahko navezuje na sliko, a je obenem zelo generično ali ohlapno.
- 3: Vprašanje je pogojno relevantno, vendar vsebuje tudi elemente, ki ob dani sliki niso povsem smiselni.
- 2: Vprašanje je precej nerelevantno in ne poizveduje po pomembnih specifikah slike.
- 1: Vprašanje je nesmiselno oziroma povsem nerelevantno za sliko, morda tudi zavajajoče.

II. Ocena kakovosti referenčnega odgovora (ocena od 1 do 5):

Ocenite referenčni odgovor ob upoštevanju spodnjih dimenzij.

1. IZPOLNITEV NALOGE (fulfilment)

- 0: zgreši bistvo ali odgovori na drugo temo;
- 1: delno; manjkajo ključni deli ali je preveč ohlapen;
- 2: v celoti odgovori na zahtevo.

2. VIZUALNA UTEMELJENOST & SPECIFIČNOST (visual_grounding)

- 0: generičen opis, sklicevanje na nevidne stvari;
- 1: nekaj konkretnih sklicev na vidne elemente;
- 2: jasne, specifične podrobnosti (položaj, velikost, barve, ostrina, prekrivanje).

3. RAZLAGA & KOHERENCA (explanation)

- 0: nelogično, protislovno, skokovito;
- 1: razumljivo, a tanko ali delno nedosledno;
- 2: jasna, povezovalna razlaga, ki podpira trditve.

4. NADZOR HALUCINACIJ (hallucination_control)

- 0: izmišljuje si objekte/odnose/barve, ki niso na sliki;
- 1: manjše pretiravanje ali ugibanje brez jasne označitve;
- 2: brez halucinacij; negotovost je ustrezno označena.

III. Izbira najboljšega odgovora modela:

- Preberi vseh 5 odgovorov modelov.
- Uporabi 4 dimenzije in sliko.

- Izberi en odgovor kot objektivno najboljši kompromis med:
- izpolnitvijo naloge, vizualno utemeljenostjo, koherenco in minimalnimi halucinacijami.

P.2 English Translation

YOUR TASK:

Look at the image, read the question and the reference answer. Below are 5 candidate answers written by different models. You have three tasks:

I. Rate the relevance of the question to the image (score from 1 to 5):

- 5: The question is relevant, clearly connected to the image, and asks about specific details or features that can be observed in the image.
- 4: The question is partially relevant and can generally be related to the image, but it is also very generic or loosely connected.
- 3: The question is conditionally relevant, but it also contains elements that do not fully make sense given the image.
- 2: The question is largely irrelevant and does not ask about important specifics of the image.
- 1: The question is nonsensical or completely irrelevant to the image, and may even be misleading.

II. Rate the quality of the reference answer (score from 1 to 5):

Evaluate the reference answer based on the dimensions below.

1. Task fulfilment
 - 0 - Misses the main point or addresses a different topic.
 - 1 - Partial; key elements are missing or treated too vaguely.
 - 2 - Fully addresses the requirements of the question.
2. Visual grounding & specificity
 - 0 - Generic description or references to things that are not visible.
 - 1 - Some concrete references to visible elements.
 - 2 - Clear, specific details (position, size, colours, overlap, composition, etc.).
3. Explanation & coherence
 - 0 - Illogical, contradictory, or disjointed.
 - 1 - Understandable but superficial or partly inconsistent.
 - 2 - A clear, coherent explanation that supports its claims.
4. Hallucination control
 - 0 - Invents major objects, relationships, or colours (major hallucination).
 - 1 - Minor exaggeration or speculation without explicitly indicating uncertainty.
 - 2 - No hallucinations about the image; uncertainty is clearly indicated where appropriate.

III. Select the best model answer:

- * Read all 5 model answers.
- * Use the 4 dimensions above together with the image.
- * Select one answer as the objectively best overall compromise between:

task fulfilment, visual grounding, coherence, and minimal hallucination.

Q Supervised Fine-Tuning Instruction

Q.1 Original Slovenian Instruction

Odgovori v slovenščini na vprašanje o sliki.
Odgovori jasno in stvarno.

Q.2 English Translation

Answer the question about the image in Slovenian.
Answer clearly and factually.

R Qualitative Base-Adapter Example

This section presents the complete responses for the Slovenian cultural- relevance question associated with image shown below. The Slovenian responses are reproduced without linguistic or factual corrections; only their original Markdown formatting has been converted to $\text{\LaTeX}$. The English versions are author translations intended to make the example accessible to non-Slovenian readers.



R.1 Slovenian Cultural Relevance (SLO) Question (SLO).

Kaj nam ta slika pove o Sloveniji?

Silver-standard answer.

Slika prikazuje slovensko in ljubljansko zastavo, ki plapolata nad grajskim stolpom v

Ljubljani, kar simbolizira nacionalno identiteto, ponos ter povezavo med državo in njenim glavnim mestom. Prisotnost obeh zastav poudarja tako državno kot tudi lokalno pripadnost.

Gemma-3-12B-IT (base model).

Slika nam pove, da se nahajamo v Sloveniji. To nam povejo naslednje stvari:

- **Slovenska zastava:** Na sliki je slovenska zastava, ki je sestavljena iz treh vodoriznih pasov - bele, modre in rdečega.
- **Madžarska zastava:** Ob slovenski zastavi je tudi madžarska zastava. To kaže na zgodovinsko povezanost med obema državama, saj sta bili obe nekoč del Avstro-Ogrske monarhije.
- **Zgradba z uro:** Zgradba z uro je verjetno državna ali občinska stavba, ki je pogosto prisotna v slovenskem urbanem okolju.
- **Rdeče strehe:** Rdeče strehe so pogoste v Sloveniji.

Skupno nam slika pove, da se nahajamo v Sloveniji, ki ima bogato zgodovino in kulturo.

Gemma-3-12B-SFT (fine-tuned model).

Slika prikazuje slovensko zastavo, ki plapolna na stavbi, kar simbolizira suverenost in nacionalno identiteto države. Prisotnost urne na stavbi poudarja zgodovinsko in administrativno vlogo mesta, ki je sedež lokalne samouprave in upravljanja z občinskimi zadevami.

R.2 English Translation

Question (SLO).

What does this image tell us about Slovenia?

Silver-standard answer.

The image shows the Slovenian and Ljubljana flags flying above the castle tower in Ljubljana, symbolising national identity, pride, and the connection between the country and its capital. The presence of both flags emphasises both national and local belonging.

Gemma-3-12B-IT (base model).

The image tells us that we are in Slovenia. The following elements indicate this:

- **Slovenian flag:** The image contains the Slovenian flag, which consists of three horizontal stripes: white, blue, and red.
- **Hungarian flag:** Next to the Slovenian flag, there is also a Hungarian flag. This indicates a historical connection between the two countries, since both were once part of the Austro-Hungarian Monarchy.
- **Building with a clock:** The building with a clock is probably a state or municipal building of a kind commonly found in Slovenian urban environments.
- **Red roofs:** Red roofs are common in Slovenia.

Overall, the image tells us that we are in Slovenia, a country with a rich history and culture.

Gemma-3-12B-SFT (fine-tuned model).

The image shows the Slovenian flag flying on a building, symbolising the country's sovereignty and national identity. The presence of the clock on the building emphasises the city's historical and administrative role as the seat of local government and the administration of municipal affairs.

R.2.1 Evaluation Outcomes

In the original five-model evaluation, the Gemma-3-12B-IT response received 2/8: one point for task fulfilment, one point for visual grounding, and zero points for both explanatory coherence and hallucination control. It was labelled as containing a major hallucination and was not selected as the best answer.

The same base-model response was evaluated again in the separate blind base–adapter comparison. In that evaluation, it received 0/8, whereas the fine-tuned response received 5/8 and was selected as the better response. The fine-tuned response received one point each for task fulfilment, visual grounding, and explanatory coherence, and two points for hallucination control. The base response was labelled as containing a major hallucination, while the fine-tuned response was not.

The difference between the two scores assigned to the identical base response reflects variation between separate judging contexts. The 2/8 score belongs to the original five-model evaluation, whereas the 0/8 and 5/8 scores belong to the dedicated blind base–adapter comparison. Consequently,

only the latter two scores should be used for the direct comparison between the base and fine-tuned models.

R.3 Foundational Example: Foreground–Background Salience (F1)

R.3.1 Original Slovenian

Question.

Opiši glavni motiv slike in elemente, ki ga podpirajo v ozadju.

Silver-standard answer.

Glavni motiv slike sta slovenska in ljubljanska zastava, ki v vetru ponosno plapolata. V ozadju in kot podlaga jima služi grajski stolp z uro. Kompozicijo dopolnjuje tudi jasno modro nebo, proti kateremu se zastavi dvigata, in delno vidno zelenje dreves.

Gemma-3-12B-IT (base model).

Glavni motiv slike sta dve zaporki, ki plapolata na stolpu.

- **Slovanska zaporka:** Na levi strani je zaporka, ki predstavlja državo Slovaško. Ima tri vodoravne pasove: belo, modro in rdečo.
- **Madžarska zaporka:** Na desni strani je zaporka, ki predstavlja državo Madžarsko. Ima tri vodoravne pasove: rdečo, belo in zeleno.

Elementi v ozadju:

- **Stolp:** Zaporki so postavljeni na stolp z uro. Stolp ima arhitekturni slog z okni in okrasnimi elementi.
- **Streha:** Pod stolpom je vidna rdeča streha zgrajene zgradbe.
- **Drevesa:** Okoli stolpa in zgradbe rastejo drevesa z zelenimi listi, ki dodajajo naravni element v sliko.
- **Nebesa:** Na ozadju je jasno modro nebo.

Gemma-3-12B-SFT (fine-tuned model).

Glavni motiv slike sta dve zastavi, ki plapolata na stolpu. Na levi plapolata slovenska zastava, na desni pa zastava mesta Ljubljana. V ozadju, ki podpira ta motiv, je jasno modro nebo, delno zakrito z zelenjem dreves v spodnjem levem kotu. Vidna je tudi del strehe z rdečo opečno kritino in del fasade stolpa z uro, ki kaže čas.

R.3.2 English Translation

Question.

Describe the main motif of the image and the background elements that support it.

Silver-standard answer.

The main motif consists of the Slovenian and Ljubljana flags, proudly flying in the wind. The castle tower with a clock serves as their background and support. The composition is further complemented by the clear blue sky, towards which the flags rise, and the partially visible foliage of the trees.

Gemma-3-12B-IT (base model).

The main motif consists of two flags flying on the tower.

- **Slavic flag:** On the left is a flag representing Slovakia. It has three horizontal stripes: white, blue, and red.
- **Hungarian flag:** On the right is a flag representing Hungary. It has three horizontal stripes: red, white, and green.

Background elements:

- **Tower:** The flags are positioned on a clock tower. The tower has windows and decorative architectural elements.
- **Roof:** A building with a red roof is visible beneath the tower.
- **Trees:** Trees with green leaves grow around the tower and the building, adding a natural element to the image.
- **Sky:** A clear blue sky is visible in the background.

Gemma-3-12B-SFT (fine-tuned model).

The main motif consists of two flags flying on the tower. The Slovenian flag is on the left, while the flag of the City of Ljubljana is on the right. The supporting background consists of a clear blue sky, partly obscured by green foliage in the lower-left corner. Part of a red-tiled roof and part of the clock-tower facade are also visible.

R.3.3 Blind-Judge Outcome

Gemini-2.5-Pro selected the fine-tuned response. The base response received 2/8: one point for task fulfilment, one for visual grounding, and zero points for both explanatory coherence and hallucination control. It was labelled as containing a major hallucination. The fine-tuned response received 8/8 and was not labelled as containing a major hallucination.

R.4 Intermediate Example: Metaphor and Visual Tropes (I3)

R.4.1 Original Slovenian

Question.

Kako se konkretni motivi na sliki spreminjajo v ideje ali čustva na simbolni ravni?

Silver-standard answer.

Zastavi, slovenska in ljubljanska, simbolizirata nacionalno in lokalno identiteto ter ponos na državo in njeno prestolnico. Njuno plapolanje visoko nad grajskim stolpom proti modremu nebu izraža občutke svobode, neodvisnosti in vitalnosti. Grajski stolp, kot zgodovinski objekt, dodaja motivu trajnost, tradicijo in vlogo zaščitnika, hkrati pa poudarja močno povezanost s preteklostjo in dediščino.

Gemma-3-12B-IT (base model).

Slika prikazuje dve zastavi, slovensko in madžarsko, ter stolp z uro. Tukaj je razčlenitev, kako se ti elementi lahko interpretirajo na simbolni ravni:

1. Zastave

- **Slovenska zastava (beli, rdeči, modri):**
 - **Bela:** Simbolizira mir, čistočo, poštenost in slovensko kulturo.
 - **Rdeča:** Predstavlja pogum, hrabrost, gore in bojovita zgodovina.
 - **Modra:** Simbolizira zvestobo, lojalnost, nebesa in slovensko morje.
 - **Na simbolni ravni:** Slovenska zastava predstavlja slovenski narod, njegovo identiteto, kulturo, zgodovino in aspiracije.
- **Madžarska zastava (beli, rdeči, zelen):**
 - **Bela:** Simbolizira mir, čistočo in upanje.
 - **Rdeča:** Predstavlja moč, hrabrost in življenje.
 - **Zelena:** Simbolizira sveto madžarsko ravnico (Alföld), plodnost in upanje.
 - **Na simbolni ravni:** Madžarska zastava predstavlja madžarski narod, njegovo identiteto, kulturo, zgodovino in vezi z deželo.

2. Stolp z uro

- **Stolp:** Pogosto simbolizira moč, stabilnost, nadzor in zgodovino.
- **Ura:** Simbolizira čas, preteklost, sedanjost in prihodnost. Pripomene na pomembnost trenutka in teka časa.
- **Na simbolni ravni:** Stolp z uro skupaj predstavlja zgodovinsko ozadje, pomembnost časa in trajnost.

3. Kombinacija vseh elementov

- **Dve zastavi na istem stolpu:** To lahko simbolizira sožitje, sodelovanje, prijateljstvo in skupno zgodovino med Slovenijo in Madžarsko. Pripomene na pretekle politične in kulturne vezi med državama.
- **Zastave in stolp z uro:** Skupaj ustvarjajo podobo, ki govori o pomembnosti zgodovine, kulture in skupnih vrednotah med državama. Pripomene na pomembnost dialoga, razumevanja in sodelovanja za prihodnost.
- **Zelenje v ozadju:** Pripomene na naravo, življenje in rast.

Čustva in ideje:

- **Pripadnost:** Pripadnost nacionalni identiteti in kulturi.
- **Prijateljstvo:** Prijateljstvo in sodelovanje med narodi.
- **Zgodovina:** Pomembnost zgodovine in spomin na preteklost.
- **Upanje:** Upanje na boljšo prihodnost in skupno usodo.
- **Mir:** Želja po miru in harmoniji.

Pomembno je poudariti, da je interpretacija simbolov subjektivna in odvisna od osebne izkušnje in kulturnega ozadja gledalca.

Gemma-3-12B-SFT (fine-tuned model).

Slovenska zastava na sliki simbolizira nacionalno identiteto, suverenost in pripadnost državi. Mestna zastava, ki je prav tako prisotna, predstavlja lokalno avtonomijo in specifično identiteto mesta. Ura na stolpu simbolizira čas, kontinuiteto in organiziranost, medtem ko celotna podoba stavbe in zastav poudarja pomen lokalne samouprave in njeno vlogo v družbi.

R.4.2 English Translation

Question.

How are the concrete motifs in the image transformed into ideas or emotions at the symbolic level?

Silver-standard answer.

The Slovenian and Ljubljana flags symbolise national and local identity and pride in the country and its capital. Their flight high above the castle tower against the blue sky conveys feelings of freedom, independence, and vitality. As a historical structure, the castle tower adds a sense of permanence, tradition, and protection while emphasising a strong connection with the past and cultural heritage.

Gemma-3-12B-IT (base model).

The image shows two flags, Slovenian and Hungarian, and a clock tower. The elements can be interpreted symbolically as follows:

1. Flags

- **Slovenian flag (white, red, and blue):**
 - **White** symbolises peace, purity, honesty, and Slovenian culture.
 - **Red** represents courage, bravery, mountains, and a martial history.
 - **Blue** symbolises loyalty, faithfulness, the sky, and the Slovenian sea.
 - **At the symbolic level**, the Slovenian flag represents the Slovenian nation, its identity, culture, history, and aspirations.
- **Hungarian flag (white, red, and green):**
 - **White** symbolises peace, purity, and hope.
 - **Red** represents strength, bravery, and life.
 - **Green** symbolises the sacred Hungarian plain (Alföld), fertility, and hope.
 - **At the symbolic level**, the Hungarian flag represents the Hungarian nation, its identity, culture, history, and connection with the land.

2. Clock tower

- The **tower** often symbolises power, stability, control, and history.
- The **clock** symbolises time, the past, present, and future, and serves as a reminder of the importance of the present moment and the passage of time.

- At the symbolic level, the clock tower represents historical background, the importance of time, and permanence.

3. Combination of the elements

- The two flags on the same tower may symbolise coexistence, cooperation, friendship, and a shared history between Slovenia and Hungary. They evoke past political and cultural ties between the two countries.
- Together, the flags and clock tower create an image about the importance of history, culture, and values shared by the two countries. They evoke the importance of dialogue, understanding, and cooperation for the future.
- The greenery in the background evokes nature, life, and growth.

The image may evoke feelings and ideas of belonging, friendship, history, hope, and peace. It is important to emphasise that the interpretation of symbols is subjective and depends on the viewer's personal experience and cultural background.

Gemma-3-12B-SFT (fine-tuned model).

The Slovenian flag in the image symbolises national identity, sovereignty, and belonging to the country. The municipal flag, which is also present, represents local autonomy and the city's specific identity. The clock on the tower symbolises time, continuity, and organisation, while the overall image of the building and flags emphasises the importance of local government and its role in society.

R.4.3 Blind-Judge Outcome

Gemini-2.5-Pro selected the fine-tuned response. The base response received 2/8: one point for task fulfilment, zero for visual grounding, one for explanatory coherence, and zero for hallucination control. It was labelled as containing a major hallucination. The fine-tuned response received 8/8 and was not labelled as containing a major hallucination.

R.5 Advanced Example: Intertextual Understanding (A2)

R.5.1 Original Slovenian

Question.

Kateri deli slike te spomnijo na druge znane podobe, prizore ali zgodbe?

Silver-standard answer.

Plapolajoči zastavi, ki se dvigata nad grajskim stolpom, močno spominjata na podobe nacionalnega ponosa in zgodovinske pomembnosti. Ta prizor lahko evokira spomine na državne proslave, zgodovinske trenutke, ko so zastave simbolizirale pomembne dogodke, ali na splošne upodobitve gradov in utrd kot čuvarjev identitete in dediščine. Gre za pogost motiv, ki povezuje preteklost s sedanostjo in poudarja občutek pripadnosti.

Gemma-3-12B-IT (base model).

Slika me spominja na več znanih podob, prizorov in zgodb:

- **Slovanska zastava:** Slovanska zastava, ki je na sliki, je simbol slovanskih narodov in se pogosto uporablja pri prireditvah, ki poudarjajo skupno slovansko identiteto.
- **Madžarska zastava:** Madžarska zastava je prepoznavna in se pogosto povezuje z Madžarsko in madžarsko kulturo.
- **Gradovi in stolpi:** Arhitektura stolpa z uro in obzidjem me spominja na evropske gradove in mestne stolpe, ki so pogosto simbol moči, zgodovine in kulture.
- **Romantični prizori:** Kombinacija zastav, stolpa in modrega neba ustvarja romantičen in idealiziran prizor, ki me spominja na slike iz zgodovinskih romanov ali filmskih prizorov.
- **Evropska identiteta:** Slika, ki prikazuje zastavi dveh slovanskih narodov, lahko simbolizira evropsko identiteto in sodelovanje med različnimi narodi.

Gemma-3-12B-SFT (fine-tuned model).

Slika, ki prikazuje Ljubljanski grad z zastavami in uro, spominja na podobe drugih zgodovinskih utrd in mestnih simbolov po Evropi. Zastave, še posebej slovenska, vzbujajo občutek nacionalne identitete in državnosti, kar je pogosto povezano z zgodovinskimi prizori in bitkami za neodvisnost.

R.5.2 English Translation**Question.**

Which parts of the image remind you of other familiar images, scenes, or stories?

Silver-standard answer.

The two flying flags rising above the castle tower strongly evoke images of national pride and historical importance. The scene may recall national celebrations, historical moments in which flags symbolised important events, or general representations of castles and fortifications as guardians of identity and heritage. It is a familiar motif that connects the past with the present and emphasises a sense of belonging.

Gemma-3-12B-IT (base model).

The image reminds me of several familiar images, scenes, and stories:

- **Slavic flag:** The Slavic flag shown in the image is a symbol of the Slavic peoples and is often used at events emphasising a shared Slavic identity.
- **Hungarian flag:** The Hungarian flag is recognisable and is often associated with Hungary and Hungarian culture.
- **Castles and towers:** The architecture of the clock tower and fortifications reminds me of European castles and city towers, which often symbolise power, history, and culture.
- **Romantic scenes:** The combination of flags, tower, and blue sky creates a romantic and idealised scene reminiscent of images from historical novels or films.
- **European identity:** By showing the flags of two Slavic nations, the image may symbolise European identity and cooperation among different nations.

Gemma-3-12B-SFT (fine-tuned model).

The image, which shows Ljubljana Castle with flags and a clock, recalls images of other historical fortifications and municipal symbols across Europe. The flags, particularly the Slovenian flag, evoke a sense of national identity and statehood, which is often associated with historical scenes and struggles for independence.

R.5.3 Blind-Judge Outcome

Gemini-2.5-Pro selected the fine-tuned response. The base response received 2/8: one point for task fulfilment, zero for visual grounding, one for explanatory coherence, and zero for hallucination control. It was labelled as containing a major hallucination. The fine-tuned response received 8/8

and was not labelled as containing a major hallucination.

These examples were selected to illustrate the progression from foundational visual description to symbolic and intertextual interpretation, as well as a recurring model-error pattern. They should not be interpreted as a representative sample of all responses. Quantitative conclusions are based on the complete evaluation.

References

- Panos Achlioptas, Maks Ovsjanikov, Kilichbek Haydarov, Mohamed Elhoseiny, and Leonidas Guibas. 2021. [ArtEmis: Affective language for visual art](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11569–11579.
- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikołaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. 2022. [Flamingo: A visual language model for few-shot learning](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 23716–23736.
- Daniel Chandler. 2022. *Semiotics: The Basics*, 4 edition. Routledge, Abingdon and New York.
- Cheng-Han Chiang and Hung-yi Lee. 2023. [Can large language models be an alternative to human evaluations?](#) In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15607–15631, Toronto, Canada. Association for Computational Linguistics.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient finetuning of quantized LLMs](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 10088–10115.
- Gemma Team. 2025. [Gemma 3 technical report](#). *arXiv preprint arXiv:2503.19786*.
- Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. 2017. [Making the v in VQA matter: Elevating the role of image understanding in visual question answering](#). In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6325–6334.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations*.
- Zaeem Hussain, Mingda Zhang, Xiaozhong Zhang, Keren Ye, Christopher Thomas, Zuha Agha, Nathan Ong, and Adriana Kovashka. 2017. [Automatic understanding of image and video advertisements](#). In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1100–1110.
- Gunther Kress and Theo van Leeuwen. 2021. *Reading Images: The Grammar of Visual Design*, 3 edition. Routledge, London and New York.
- Julia Kruk, Caleb Ziems, and Diyi Yang. 2023. [Impressions: Visual semiotics and aesthetic impact understanding](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12273–12291, Singapore. Association for Computational Linguistics.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. [Visual instruction tuning](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 34892–34916.
- Matej Martinc and Domen Vreš. 2026. [A large-scale instruction-tuning dataset and models for Slovenian vision-language tasks](#). In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 9420–9433, Palma de Mallorca, Spain. ELRA Language Resource Association.
- Lia Morra, Antonio Santangelo, Pietro Basci, Luca Piano, Fabio Garcea, Fabrizio Lamberti, and Massimo Leone. 2024. For a semiotic ai: Bridging computer vision and visual semiotics for computational observation of large scale facial image archives. *Computer Vision and Image Understanding*, 249:104187.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR.

Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. 2019. [Towards VQA models that can read](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8317–8326.

Hartmut Stöckl. 2024. [Bold and impactful: A reappraisal of gunther kress’s \(social\) semiotic legacy in the light of current multimodality research](#). *Text & Talk*, 44(4):469–491.

Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. 2024. [Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution](#). *arXiv preprint arXiv:2409.12191*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging LLM-as-a-judge with MT-Bench and chatbot arena](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623.